

Transfer Attacks in Face Recognition

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Attack Considered

- LI_BOOST_MI

Why LI_BOOST_MI?

- LI_BOOST_MI because it is a recent (2023) transferability enhancement technique that builds upon the strong MI-FGSM baseline.
- Traditional transfer attacks often overfit to the surrogate model, reducing their effectiveness on unseen victim models.
- LI-Boost addresses this issue by introducing local invariance through multiple locally transformed samples and aggregated gradients.
- This helps generate more robust and transferable adversarial perturbations while remaining compatible with the existing attack generation, evaluation, and analysis pipeline.

Repository link :-

<https://github.com/Charushi06/transferattackInterns>

Definition & Nature

- White-box adversarial attack
- Gradient-based optimization method
- Iterative perturbation refinement
- L_∞ -bounded constrained attack
- Momentum-based attack enhanced with Local Invariance Boosting (LI-Boost)
- Transferability enhancement attack

LI_BOOST_MI - Goal

- Operates in embedding space
- Impersonation: increase cosine similarity between adversarial image and target embedding
- Dodging: decrease cosine similarity between adversarial image and target embedding
- Improves transferability across unseen victim models
- Reduces overfitting to the surrogate (attacker) model by enforcing local invariance

Basic Idea

- LI_BOOST_MI extends MI-FGSM by introducing Local Invariance Boosting.
- Instead of computing gradients from a single image at each iteration:
- Generate multiple locally translated versions of the current adversarial image
- Compute gradients for each translated sample
- Aggregate the gradients
- Apply momentum accumulation
- Update the adversarial image using the aggregated direction
- This encourages the perturbation to learn model-agnostic features that transfer better to unseen models.

Attack Implementation

Working of LI_BOOST_MI

- Initialize adversarial image as the original input image.
- Compute the target embedding.
- For every iteration:
 - Generate multiple translated versions of the current adversarial image.
 - Compute embeddings for each translated sample.
 - Calculate cosine similarity with the target embedding.
 - Compute gradients for all translated samples.
 - Aggregate the gradients.
 - Normalize the gradients.
 - Update momentum.
 - Update the adversarial image using the sign of the momentum.
 - Project the image back into the ϵ -bounded region.
 - Clip pixel values to the valid range.
- Return the final adversarial image.

Implementation

- Built on top of MI-FGSM
- Uses ArcFace as the surrogate attacker model
- Uses cosine similarity as the attack objective
- Applies Local Invariance through translated image samples
- Aggregates gradients from multiple local transformations
- Uses momentum accumulation:

$$g_t = \mu g_{t-1} + \nabla J$$

- Uses sign-based iterative updates
- Uses step size:

$$\alpha = \frac{\epsilon}{NUM_ITER}$$

- Enforces explicit L_∞ constraints using clipping:

$$x - \epsilon \leq x_{adv} \leq x + \epsilon$$

- Clips pixel values to the valid normalized range $[-1,1]$
- Evaluated using cosine similarity against multiple victim face recognition models

Result

Overall Benchmark Results

Attack Method	Attacker-Victim Evaluations	Breach Rate (%)	Mean Embedding Impact
SI_NI_FGSM	480	31.46	0.1788
MI_FGSM	480	26.04	0.1618
MI_ADMIX_DI_TI	480	23.96	0.1509
TI_FGSM	480	20.00	0.1290
PGD	480	16.46	0.1011

Controlled ArcFace-Based Comparison

Attack Method	Attacker-Victim Evaluations	Breach Rate (%)	Mean Embedding Impact
LI_BOOST_MI (New)	90	38.89	0.1900
SI_NI_FGSM	90	34.44	0.1825
MI_ADMIX_DI_TI	90	25.56	0.1339
MI_FGSM	90	24.44	0.1502
TI_FGSM	90	18.89	0.1161
MI_BSR	90	16.67	0.0974
PGD	90	11.11	0.0748

Conclusion

LI_BOOST_MI achieved the highest breach rate of 38.89%, outperforming all baseline attacks. Compared to the strongest baseline, SI_NI_FGSM (34.44%), LI_BOOST_MI achieved an absolute improvement of 4.45 percentage points while also producing the highest mean embedding impact (0.1900).

Thank You

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